FURTHER MECHANICS / CENTRE OF MASS

Centre of Mass of Solids of Revolution

Mark scheme 11 questions · 123 marks

EXAM BOARDS Edexcel FM2 9FM0/4C · AQA 7367/3M · OCR A Y543 · OCR MEI Mech Major Y421

Questions

Question	Marks by part	Marks	Page
Question 1		8	4
Question 2	(a) 6 (b) 3	9	5
Question 3	(a) 1 (b) 8 (c) 3	12	6
Question 4	(a) 4 (b) 7	11	7
Question 5	(a) 8 (b) 3	11	8
Question 6	(a) 1 (b) 6 (c)(i) 3 (c)(ii) 4	14	9
Question 7	(a) 4 (b) 6	10	11
Question 8	(a) 4 (b) 8	12	12
Question 9	(a) 4 (b) 8	12	13
Question 10	(a) 8 (b) 3	11	14
Question 11	(a) 6 (b) 7	13	15
	Total	**123**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes. A grey line gives the rule where the name alone doesn't.

ft	follow through Correct work from the student's own earlier wrong value earns the mark.	**AG**	answer given The answer is printed in the question, so every step to it must be shown.
cao	correct answer only No follow-through from an earlier error.	**cso**	correct solution only No error anywhere in the part, not just in its answer.
oe	or equivalent	**awrt**	answers which round to
soi	seen or implied Earned if written, or if later working shows it was used.	**isw**	ignore subsequent working A correct answer, once reached, isn't spoiled by later working.
SC	special case Credit, as the note says, for work outside the main method.	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- **[*n*]** ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1	$(V =)\ \pi \int_0^2 \left(16 - 8x^2 + x^4\right) \mathrm{d}x$ $= \pi \left[16x - \frac{8x^3}{3} + \frac{x^5}{5}\right]_0^2$	**M1 A1**	**M1** Expands y^2 and integrates $\pi \int y^2 \, \mathrm{d}x$, at least one power increased by 1. Condone π omitted from both integrals. **A1** Correct integrated expression; limits not needed, and π may be omitted.
	$V = \frac{256\pi}{15}$	**A1**	oe, or $\frac{256}{15}$ with π omitted from both integrals.
	$\pi \int_0^2 \left(16x - 8x^3 + x^5\right) \mathrm{d}x$ $= \pi \left[8x^2 - 2x^4 + \frac{x^6}{6}\right]_0^2$	**M1 A1**	**M1** Integrates $\pi \int xy^2 \, \mathrm{d}x$, at least one power increased by 1. Condone π omitted from both integrals. **A1** Correct integrated expression; limits not needed, and π may be omitted.
	$V\bar{x} = \frac{32\pi}{3}$	**A1**	oe, or $\frac{32}{3}$ with π omitted from both integrals.
	$\bar{x} = \text{“}\frac{32\pi}{3}\text{”} \div \text{“}\frac{256\pi}{15}\text{”} = \ldots$	**dM1**	Divides their moment by their volume, both found by integration. Not for volume divided by moment.
	$\bar{x} = \frac{5}{8}$	**A1**	$\frac{5}{8}$ or 0.625.
Total for Question 1		**8**	

Question	Scheme	Marks	Notes
2(a)	$V = \pi \int_1^4 \frac{1}{x}\,\mathrm{d}x = \pi\left[\ln x\right]_1^4$	**M1**	Uses $\pi \int y^2\,\mathrm{d}x$ with y^2 in terms of x, integrating to the form $k\ln x$. Condone π omitted from both integrals.
	$V = 2\pi\ln 2$	**A1**	oe such as $\pi\ln 4$, or $2\ln 2$ with π omitted from both integrals.
	$V\bar{x} = \pi \int_1^4 x \times \frac{1}{x}\,\mathrm{d}x = \pi\left[x\right]_1^4$	**M1**	Uses $\pi \int xy^2\,\mathrm{d}x$ with y^2 in terms of x and integrates. Condone π omitted from both integrals.
	$V\bar{x} = 3\pi$	**A1**	Or 3 with π omitted from both integrals.
	$\bar{x} = \frac{\text{“}3\pi\text{”}}{\text{“}2\pi\ln 2\text{”}} \Rightarrow \text{distance} = \bar{x} - 1$	**dM1**	Divides their moment by their volume and subtracts 1 for the base at $x = 1$.
	$\frac{3}{2\ln 2} - 1$	**A1 AG**	Correct volume and moment, with π in both or neither, divided and with 1 subtracted, leading without error to the printed distance.
		[6]	
2(b)	$\tan\alpha^\circ = \frac{1}{\frac{3}{2\ln 2} - 1}$	**M1**	Forms $\tan\alpha$ from the radius 1 and the distance of the centre of mass from the base. Condone the ratio inverted.
	$\tan\alpha^\circ = \frac{2\ln 2}{3 - 2\ln 2}$ $[\approx 0.859]$	**A1**	Ratio the right way up; any equivalent form or awrt 0.859.
	$\alpha = 40.7$	**A1**	awrt 40.7
		[3]	
Total for Question 2		**9**	

Question	Scheme	Marks	Notes
3(a)	R uniform and symmetric about the x-axis $\Rightarrow G$ on the x-axis, so $y = 0$	**E1**	States that R is symmetric about the x-axis, so G lies on it. Not for "uniform" alone.
		[1]	
3(b)	$V = \pi \int_1^3 \left(x^3 - 3x^2 + 4x\right) \mathrm{d}x$ $= \pi \left[\frac{x^4}{4} - x^3 + 2x^2\right]_1^3$	**M1 A1**	**M1** Integrates y^2, power of at least one term increased by 1. Condone π omitted from both integrals. **A1** Correct integral; π may be omitted.
	$V = 10\pi$	**A1**	Or 10 with π omitted from both integrals.
	$\pi \int_1^3 \left(x^4 - 3x^3 + 4x^2\right) \mathrm{d}x$ $= \pi \left[\frac{x^5}{5} - \frac{3x^4}{4} + \frac{4x^3}{3}\right]_1^3$	**M1 A1**	**M1** Multiplies y^2 by x and integrates, power of at least one term increased by 1. Condone π omitted from both integrals. **A1** Correct integral; π may be omitted.
	$= \frac{346\pi}{15}$	**A1**	oe, or $\frac{346}{15}$ with π omitted from both integrals.
	$\bar{x} = \frac{\text{"}346\pi/15\text{"}}{\text{"}10\pi\text{"}} = \ldots$	**dM1**	Divides their moment by their volume, each from an attempted integration.
	$\bar{x} = \frac{173}{75}$	**A1**	Any exact equivalent fraction, such as $\frac{346}{150}$.
		[8]	
3(c)	G below P: $\tan\theta = \frac{\sqrt{2}}{\text{"}\frac{173}{75}\text{"} - 1}$ $\left[= \frac{75\sqrt{2}}{98}\right]$	**M1 A1ft**	**M1** Forms a trigonometric ratio for the angle from the right-angled triangle with sides their face radius and their $\bar{x} - 1$. Condone the ratio inverted. **A1ft** Correct ratio the right way up, ft their $\bar{x}$.
	$\theta = 47.3°$	**A1**	awrt 47.3°.
		[3]	
	Total for Question 3	**12**	

Question	Scheme	Marks	Notes
4(a)	$V = \pi \int_0^1 e^{2y}\,\mathrm{d}y$ $= \pi \left[\frac{1}{2}e^{2y}\right]_0^1$	**M1 A1**	**M1** Forms $\pi \int x^2\,\mathrm{d}y$ with x^2 in terms of y and integrates to the form ke^{my}. Condone a missing π. **A1** Correct integral, with or without π.
	$= \pi\left(\frac{1}{2}e^2 - \frac{1}{2}\right)$	**dM1**	Substitutes the limits into their integrated expression and subtracts.
	$= \frac{\pi}{2}\left(e^2 - 1\right)$	**A1 AG**	Correct substituted expression leading without error to the printed volume, with π present; combined steps accepted.
		[4]	
4(b)	$V\bar{y} = \pi \int_0^1 y\,e^{2y}\,\mathrm{d}y$	**M1**	Forms the moment as $\pi \int yx^2\,\mathrm{d}y$ with x^2 in terms of y. Condone π omitted from both integrals.
	$\int y\,e^{2y}\,\mathrm{d}y = \frac{1}{2}y\,e^{2y} - \int \frac{1}{2}e^{2y}\,\mathrm{d}y$	**M1**	Applies integration by parts with $u = y$, reaching the form $py\,e^{2y} - \int qe^{2y}\,\mathrm{d}y$. Condone coefficient slips.
	$= \frac{1}{2}y\,e^{2y} - \frac{1}{4}e^{2y}$	**A1**	
	$\left(\frac{1}{2}e^2 - \frac{1}{4}e^2\right) - \left(0 - \frac{1}{4}\right)$	**dM1**	Substitutes limits 0 and 1 into their expression of the form $py\,e^{2y} + qe^{2y}$ and subtracts.
	$V\bar{y} = \frac{\pi}{4}\left(e^2 + 1\right)$	**A1**	oe, or $\frac{1}{4}\left(e^2 + 1\right)$ with π omitted from both integrals.
	$\bar{y} = \dfrac{\text{“}\frac{\pi}{4}\left(e^2+1\right)\text{”}}{\frac{\pi}{2}\left(e^2-1\right)}$	**dM1**	Divides their moment by the volume from (a). Condone π omitted from both integrals.
	$\bar{y} = \dfrac{e^2+1}{2\left(e^2-1\right)}$	**A1**	Any equivalent exact form in e. A decimal alone scores A0.
		[7]	
	Total for Question 4	**11**	

Question	Scheme	Marks	Notes
5(a)	$V = \pi \int_0^2 \frac{16}{(2+x)^2}\,\mathrm{d}x$ $= 16\pi\left[-\frac{1}{2+x}\right]_0^2 = \ldots$	**M1**	Uses $\pi \int y^2\,\mathrm{d}x$ with y^2 in terms of x, integrating to the form $\frac{k}{2+x}$ and applying limits. Condone π omitted from both integrals.
	$V = 4\pi$	**A1**	Or 4 with π omitted from both integrals.
	$\frac{x}{(2+x)^2} = \frac{1}{2+x} - \frac{2}{(2+x)^2}$ $\Rightarrow \int \frac{x}{(2+x)^2}\,\mathrm{d}x = \ldots$	**M1**	Writes $\frac{x}{(2+x)^2}$ in integrable form, by partial fractions, $u = 2+x$ or parts, and integrates to $p\ln(2+x) + \frac{q}{2+x}$ oe. Condone coefficient slips.
	$\pi \int_0^2 \frac{16x}{(2+x)^2}\,\mathrm{d}x$ $= 16\pi\left[\ln(2+x) + \frac{2}{2+x}\right]_0^2$	**A1**	Correct integral oe, including $\ln u + \frac{2}{u}$ with $u = 2+x$. Modulus signs accepted; π may be omitted.
	$16\pi\left[\left(\ln 4 + \frac{1}{2}\right) - (\ln 2 + 1)\right]$	**dM1**	Substitutes the limits 2 and 0, or 4 and 2 in u, into their integral and subtracts.
	$= 16\pi\left(\ln 2 - \frac{1}{2}\right)$	**A1**	oe, such as $\pi(16\ln 2 - 8)$, or $16\ln 2 - 8$ with π omitted from both integrals.
	$\bar{x} = \frac{\text{“}16\pi\left(\ln 2 - \frac{1}{2}\right)\text{”}}{\text{“}4\pi\text{”}}$ $[= 4\ln 2 - 2]$ $2 - \bar{x} = \ldots$	**dM1**	Divides their moment by their volume and subtracts from 2, the x-coordinate of the smaller face.
	$2 - (4\ln 2 - 2) = 4(1 - \ln 2)$	**A1 AG**	Correct volume and moment, with π in both or neither, leading without error to $4(1-\ln 2)$, with the smaller face at $x = 2$ soi.
		[8]	
5(b)	G vertically below P: $\tan\theta = \frac{1}{4(1-\ln 2)}$	**M1 A1**	**M1** Uses a right-angled triangle with sides 1 and the distance from (a), forming a trigonometric ratio for the angle. Condone the ratio inverted. **A1** Ratio the right way up.
	$\theta = 39°$	**A1**	
		[3]	
	Total for Question 5	**11**	

Question	Scheme	Marks	Notes
6(a)	The solid is uniform (constant density)	**B1**	
		[1]	
6(b)	$V = \pi \int_0^9 4x \, \mathrm{d}x = \pi \left[2x^2\right]_0^9$	**M1**	Uses $\pi \int y^2 \, \mathrm{d}x$ with y^2 in terms of x, increasing the power by 1. Condone π omitted from both integrals.
	$V = 162\pi$	**A1**	Or 162 with π omitted from both integrals.
	$\pi \int_0^9 4x^2 \, \mathrm{d}x = \pi \left[\frac{4}{3}x^3\right]_0^9$	**M1**	Uses $\pi \int xy^2 \, \mathrm{d}x$ with y^2 in terms of x, increasing the power by 1. Condone π omitted from both integrals.
	$\pi \int_0^9 xy^2 \, \mathrm{d}x = 972\pi$	**A1**	Or 972 with π omitted from both integrals.
	$\bar{x} = \frac{\text{“}972\pi\text{”}}{\text{“}162\pi\text{”}} = \ldots \Rightarrow 9 - \bar{x} = \ldots$	**dM1**	Divides their moment by their volume and finds the distance from the plane face at $x = 9$.
	$\bar{x} = 6 \Rightarrow 9 - 6 = 3\,\mathrm{cm}$	**A1 AG**	Correct volume and moment, with π in both or neither, leading without error to the distance 3 from the plane face.
		[6]	
6(c)(i)	Radius of plane face $= 2\sqrt{9} = 6$	**B1**	
	$\tan\theta = \frac{\text{“}6\text{”}}{3} \Rightarrow \theta = \ldots$	**M1**	Uses a tangent with their radius and 3. Condone the ratio inverted.
	$\theta = 63°$	**A1**	awrt 63°.
		[3]	
6(c)(ii)	$R = mg\cos\theta,\ F = mg\sin\theta,\ F = 1.5R$	**M1**	Resolves along and perpendicular to the board and uses $F = 1.5R$, or uses $\tan\theta = \mu$ with $\mu = 1.5$. Condone sign slips.
	$\tan\theta = 1.5 \Rightarrow \theta \approx 56.3°$	**A1**	$\tan\theta = 1.5$ or awrt 56.3°.

Question 6 continued

Question	Scheme	Marks	Notes
	Compares with toppling: $56.3° < \text{“}63.4°\text{”}$ $\Rightarrow$ slides first	**dM1** **A1ft**	**dM1** Compares their sliding angle with their toppling angle from (c)(i), or 1.5 with their $\tan\theta$ for toppling. **A1ft** ft their angles or tangents. Conclusion stated with the comparison.
		[4]	
	Total for Question 6	**14**	

Question	Scheme	Marks	Notes
7(a)	$V = \frac{\pi a^2}{b^2}\int_{b/2}^{b}\left(b^2 - x^2\right)\mathrm{d}x$ $= \frac{\pi a^2}{b^2}\left[b^2x - \frac{x^3}{3}\right]_{b/2}^{b}$	**M1 A1**	**M1** Uses $\pi\int y^2\,\mathrm{d}x$ with y^2 in terms of x and integrates, raising at least one power by 1. Condone a slip in the constant factor or a missing π. **A1** Correct integral, with or without π.
	$\frac{\pi a^2}{b^2}\left[\left(b^3 - \frac{b^3}{3}\right) - \left(\frac{b^3}{2} - \frac{b^3}{24}\right)\right]$ $= \frac{5}{24}\pi a^2 b$	**dM1** **A1 AG**	**dM1** Substitutes b and $\frac{b}{2}$ into their integral and subtracts, the right way round. **A1 AG** Correct substituted expression leading without error to $\frac{5}{24}\pi a^2 b$, with π present; collected steps accepted.
		[4]	
7(b)	$\pi\int_{b/2}^{b} xy^2\,\mathrm{d}x = \frac{\pi a^2}{b^2}\int_{b/2}^{b}\left(b^2x - x^3\right)\mathrm{d}x$ $= \frac{\pi a^2}{b^2}\left[\frac{b^2x^2}{2} - \frac{x^4}{4}\right]_{b/2}^{b}$	**M1 A1**	**M1** Uses $\pi\int xy^2\,\mathrm{d}x$ with y^2 in terms of x and integrates, raising at least one power by 1. Condone a slip in the constant factor, or π omitted from both integrals. **A1** Correct integral; π may be omitted.
	$\frac{\pi a^2}{b^2}\left[\left(\frac{b^4}{2} - \frac{b^4}{4}\right) - \left(\frac{b^4}{8} - \frac{b^4}{64}\right)\right]$ $= \frac{9}{64}\pi a^2 b^2$	**dM1 A1**	**dM1** Substitutes b and $\frac{b}{2}$ into their integral and subtracts, the right way round. **A1** oe, or $\frac{9}{64}a^2b^2$ with π omitted from both integrals.
	$\bar{x} = \text{“}\frac{9}{64}\pi a^2 b^2\text{”} \div \frac{5}{24}\pi a^2 b = \frac{27b}{40}$ $\frac{27b}{40} - \frac{b}{2} = \frac{7b}{40}$	**dM1** **A1 AG**	**dM1** Divides their moment by the volume and subtracts $\frac{b}{2}$, or otherwise measures the distance from the plane face. Condone π omitted from both integrals. **A1 AG** Correct moment and volume, with π in both or neither, leading without error to $\frac{7b}{40}$; collected steps accepted.
		[6]	
Total for Question 7		**10**	

Question	Scheme	Marks	Notes
8(a)	$V = \pi \int_0^{\ln 3} \left(e^{2y} - 2e^y + 1\right) \mathrm{d}y$ $= \pi \left[\frac{1}{2}e^{2y} - 2e^y + y\right]_0^{\ln 3}$	**M1 A1**	**M1** Uses x in terms of y in $\int x^2 \, \mathrm{d}y$, expands and integrates, with at least one term correct. Condone a missing π. **A1** Correct integral, with or without π.
	$\pi \left[\left(\frac{9}{2} - 6 + \ln 3\right) - \left(\frac{1}{2} - 2\right)\right]$	**dM1**	Substitutes $\ln 3$ and 0 into their integral and subtracts.
	$= \pi \left(\ln 3 - \frac{3}{2} + \frac{3}{2}\right) = \pi \ln 3$	**A1 AG**	Correct substituted expression leading without error to $\pi \ln 3$, with π present; combined steps accepted.
		[4]	
8(b)	$V\bar{y} = \pi \int_0^{\ln 3} y \left(e^{2y} - 2e^y + 1\right) \mathrm{d}y$	**M1**	Forms $\pi \int yx^2 \, \mathrm{d}y$ with x in terms of y, squared or expanded. Condone missing limits, or π omitted from both integrals.
	$\int ye^{2y} \, \mathrm{d}y = \frac{1}{2}ye^{2y} - \int \frac{1}{2}e^{2y} \, \mathrm{d}y$	**M1**	Uses integration by parts on ye^{2y} or ye^y, reaching $pye^{ky} \pm \int qe^{ky} \, \mathrm{d}y$. Condone sign and coefficient slips.
	$= \frac{1}{2}ye^{2y} - \frac{1}{4}e^{2y}$	**A1**	
	$\int ye^y \, \mathrm{d}y = ye^y - e^y$ $\int y \, \mathrm{d}y = \frac{1}{2}y^2$	**A1**	Both correct.
	$\pi \left[\left(\frac{9}{2}\ln 3 - \frac{9}{4}\right) + \frac{1}{4} - 2\left(3\ln 3 - 3 + 1\right)\right.$ $\left. + \frac{1}{2}(\ln 3)^2\right]$	**dM1**	Substitutes $\ln 3$ and 0 into their integrated terms and subtracts.
	$V\bar{y} = \pi \left(2 - \frac{3}{2}\ln 3 + \frac{1}{2}(\ln 3)^2\right)$	**A1**	Correct moment, oe, or $2 - \frac{3}{2}\ln 3 + \frac{1}{2}(\ln 3)^2$ with π omitted from both integrals.
	$\bar{y} = \frac{\text{“}\pi\left(2 - \frac{3}{2}\ln 3 + \frac{1}{2}(\ln 3)^2\right)\text{”}}{\pi \ln 3}$	**dM1**	Divides their moment by the volume $\pi \ln 3$. Condone π omitted from both integrals.
	$\bar{y} = \frac{4 - 3\ln 3 + (\ln 3)^2}{2\ln 3}$	**A1**	Any exact equivalent, such as $\frac{2}{\ln 3} - \frac{3}{2} + \frac{1}{2}\ln 3$.
		[8]	
Total for Question 8		**12**	

Question	Scheme	Marks	Notes
9(a)	$V = \pi \int_0^{\frac{\pi}{4}} \left(\sec^2 x - 1\right) \mathrm{d}x$ $= \pi \left[\tan x - x\right]_0^{\frac{\pi}{4}}$	**M1 A1**	**M1** Uses $\tan^2 x = \sec^2 x - 1$ and integrates at least one term correctly. Condone a sign slip in the identity or a missing π. **A1** Correct integral, with or without π.
	$V = \pi\left(1 - \frac{\pi}{4}\right) - \pi(0 - 0)$ $= \frac{\pi}{4}(4 - \pi)$	**dM1** **A1 AG**	**dM1** Substitutes $\frac{\pi}{4}$ and 0 into their integrated expression and subtracts. **A1 AG** Correct substituted expression leading without error to the printed result, with π present.
		[4]	
9(b)	$\pi \int_0^{\frac{\pi}{4}} x \tan^2 x \, \mathrm{d}x$ $= \pi \int_0^{\frac{\pi}{4}} \left(x \sec^2 x - x\right) \mathrm{d}x$	**M1**	Forms $\pi \int x \tan^2 x \, \mathrm{d}x$ and uses $\tan^2 x = \sec^2 x - 1$, here or within integration by parts. Condone π omitted from both integrals.
	$\int x \sec^2 x \, \mathrm{d}x = x \tan x - \int \tan x \, \mathrm{d}x$	**M1 A1**	**M1** Applies integration by parts with $u = x$, reaching $x\,g(x) - \int g(x) \, \mathrm{d}x$, g their integral of $\sec^2 x$ or $\tan^2 x$. Condone sign slips. **A1** Correct, or $x(\tan x - x) - \int (\tan x - x) \, \mathrm{d}x$.
	$\pi \left[x \tan x - \ln(\sec x) - \frac{x^2}{2}\right]_0^{\frac{\pi}{4}}$	**A1**	oe, such as $+\ln(\cos x)$ or $\ln\lvert\sec x\rvert$; π may be omitted.
	$\pi\left(\frac{\pi}{4} - \ln\sqrt{2} - \frac{\pi^2}{32}\right) - \pi(0)$	**dM1**	Substitutes $\frac{\pi}{4}$ and 0 into their integrated expression and subtracts; the zero lower-limit terms may be implied.
	$= \frac{\pi}{32}\left(8\pi - 16\ln 2 - \pi^2\right)$	**A1**	Any exact form, or $\frac{1}{32}\left(8\pi - 16\ln 2 - \pi^2\right)$ with π omitted from both integrals.
	$\bar{x} = \text{“}\frac{\pi}{32}\left(8\pi - 16\ln 2 - \pi^2\right)\text{”} \div \frac{\pi}{4}(4 - \pi)$	**dM1**	Divides their moment by the volume from (a). Condone π omitted from both integrals.
	$\bar{x} = \frac{8\pi - 16\ln 2 - \pi^2}{8(4 - \pi)}$	**A1**	Any exact equivalent.
		[8]	
Total for Question 9		**12**	

Question	Scheme	Marks	Notes
10(a)	$V_1 = \frac{1}{3}\pi e^2 \times 2 = \frac{2\pi e^2}{3}$	**B1**	Or $\frac{2e^2}{3}$ with π omitted throughout.
	$x_1 = 2 - \frac{1}{4} \times 2 = \frac{3}{2}$	**B1**	Or $\frac{1}{2}$ from the base towards O, clearly placing it at $x = \frac{3}{2}$.
	$V_2 = \pi \int_2^4 e^x \, \mathrm{d}x = \pi \left[e^x\right]_2^4 = \ldots$	**M1**	Uses $\pi \int y^2 \, \mathrm{d}x$ with y^2 in terms of x, integrates and applies limits. Condone π omitted throughout.
	$V_2 = \pi \left(e^4 - e^2\right)$	**A1**	Or $e^4 - e^2$ with π omitted throughout.
	$\int_2^4 xe^x \, \mathrm{d}x = \left[xe^x\right]_2^4 - \int_2^4 e^x \, \mathrm{d}x$	**M1**	Applies integration by parts to xe^x, reaching the form $pxe^x - q\int e^x \, \mathrm{d}x$. Condone sign and coefficient slips.
	$V_2 x_2 = \pi \left(3e^4 - e^2\right)$	**A1**	Or $3e^4 - e^2$ with π omitted throughout.
	$\left(\frac{2\pi e^2}{3} + \text{“}\pi(e^4 - e^2)\text{”}\right)\bar{x}$ $= \frac{2\pi e^2}{3} \times \frac{3}{2} + \text{“}\pi(3e^4 - e^2)\text{”}$	**M1**	Forms total volume $\times \bar{x}$ equal to the sum of each piece's volume times its centre of mass, using their values for both pieces.
	$\frac{\pi e^2(3e^2-1)}{3}\bar{x} = 3\pi e^4 \Rightarrow \bar{x} = \frac{9e^2}{3e^2-1}$	**A1 AG**	Correct substituted moments equation, with π in every term or none, leading without error to the printed result; combined simplification accepted.
		[8]	
10(b)	$\tan\theta = \frac{e^2}{4 - \frac{9e^2}{3e^2-1}}$	**M1**	Uses G vertically below B, forming the tangent of the angle from e^2 and $4 - \bar{x}$. Condone the ratio inverted.
	$\tan\theta = \frac{e^2(3e^2-1)}{3e^2-4}$ $[\approx 8.61]$	**A1**	Ratio the right way up; any exact form or awrt 8.61.
	$\theta \approx 83.4°$	**A1**	awrt 83.4°, or awrt 1.46 radians.
		[3]	
	Total for Question 10	**11**	

Question	Scheme	Marks	Notes
11(a)	$V = \pi\int_0^2 (4y)^{\frac{2}{3}}\,\mathrm{d}y$ $= 2^{\frac{4}{3}}\pi\left[\frac{3}{5}y^{\frac{5}{3}}\right]_0^2$	**M1**	Forms $\pi\int x^2\,\mathrm{d}y$ with x^2 in terms of y and integrates: power increased by 1. Limits need not be seen. Condone a missing π.
	$V = \frac{24\pi}{5}$	**A1**	Exact, oe such as 4.8π.
	$\pi\int_0^2 y(4y)^{\frac{2}{3}}\,\mathrm{d}y = 2^{\frac{4}{3}}\pi\left[\frac{3}{8}y^{\frac{8}{3}}\right]_0^2$	**M1**	Forms $\pi\int yx^2\,\mathrm{d}y$, or the moment about the plane face, with x^2 in terms of y and integrates: power increased by 1. Condone π omitted from both integrals.
	$= 6\pi$	**A1**	Or $\frac{18\pi}{5}$ about the plane face; 6 or $\frac{18}{5}$ with π omitted from both integrals.
	$\bar{y} = \frac{\text{“}6\pi\text{”}}{\text{“}\frac{24\pi}{5}\text{”}}$ $[= \frac{5}{4}] \Rightarrow 2 - \bar{y} = \ldots$	**dM1**	Divides their moment by their volume and gives the distance from the plane face $y = 2$.
	$2 - \frac{5}{4} = \frac{3}{4}\,\text{cm}$	**A1 AG**	Correct moment and volume, with π in both or neither, leading without error to $\frac{3}{4}$.
		[6]	
11(b)	$m_C = 12\pi\rho, \quad m_S = \frac{24\pi k\rho}{5}$	**B1**	Both masses, or in the correct ratio oe such as $5 : 2k$.
	$\frac{3}{2}, \quad 3 + \frac{3}{4} = \frac{15}{4}$	**B1**	Both distances, from the base or another consistent reference point.
	$\left(12 + \frac{24k}{5}\right)\bar{x} = 12 \times \frac{3}{2} + \frac{24k}{5} \times \frac{15}{4}$	**M1**	Forms a moments equation with total mass and both mass-distance terms. Condone slips in values.
	$\bar{x} = \frac{15(1+k)}{10+4k}$	**A1**	oe, from the base or consistent with their reference point.
	$\bar{x}\tan\alpha = 2 \Rightarrow \frac{4}{5}\bar{x} = 2$	**M1**	Uses $\tan\alpha = \frac{2}{\bar{x}}$ at toppling, as an equation or inequality, with their $\bar{x}$ converted to the distance from the base where needed, or a value. Condone the ratio inverted.
	$\text{“}\frac{15(1+k)}{10+4k}\text{”} = \frac{5}{2} \Rightarrow k = \ldots$	**dM1**	Equates or compares their $\bar{x}$, in terms of k, with their critical distance from their reference point and solves for k.

Question 11 continued

Question	Scheme	Marks	Notes
	$k = 2$	**A1**	cao. Accept $k \leq 2$.
		[7]	
	Total for Question 11	**13**	